

Description of learning paths and courses

This document gives a more precise description of the courses involved in the new BSc in Mathematics, and of the corresponding Learning Paths (LPs). It also describes scheduling constraints associated with the programme

This is a list of our LPs:

LP1: Algebra and Geometry

LP2: Analysis and Dynamical Systems

LP3: Probability and Statistics

LP4: Biomedical Mathematics

LP5: Computer Science

LP6: Data Science

LP7: Optimisation and Economics

Important notes:

- * Each LP is collection of courses that form a coherent learning stream.
- * LPs do not generate any constraints or progression rules for students. Constraints and progression rules of our programme are specified in the proposal, and reported also in this document, under the section **Constraints and progression rules**
- * Each LP automatically satisfies the constraints and progression rules of our programme
- * LPs are not visible on graduating certificates
- * LP combine courses from the BSc in Mathematics, and courses from other programmes. Signposted with this colours

Course taught for the BSc in
Mathematics

Course taught outside of the
BSc in Mathematics

- * Scheduling constraints for the LPs are specified below.

YEAR 1 All LPs are identical in year 1

	All Learning Paths (LP1-LP6)
Periods 1-3	Single Variable Calculus
	Basic Concepts in Mathematics
	Discrete Mathematics
	Linear Algebra
	English Language Test
	Intro to Math Modelling (and LATEX)
	Intro to Programmemeing (Python)
Periods 4-6	Probability Theory
	Mathematical Analysis
	Group Theory
	Multivariable Calculus
	Mathematical Modelling of Dynamical Systems

YEAR 2: In this year there are still many similarities among paths. Reading the table horizontally it is clear that most courses are shared between LPs.

[illegible]

Year 3: in this year LPs are more diversified. For each LP we indicate the courses on which we commit to have no clashes in the schedule. We also indicate courses that are good fit for the LP, but on which clashes may occur, so we leave it to students to manage potential clashes. We aim to have the content of this table written in the UAS.

	LP1: Algebra and Geometry	LP2: Analysis and Dynamical Systems	LP3: Probability and Statistics	LP4: Biomedical Mathematics	LP5: Computer Science	LP6: Data Science	LP7: Optimisation and Economics
Periods 1-3 no clashes in the schedule	Differential Geometry	Differential Geometry	Biomedical mathematics	Biomedical mathematics	Data structures & algorithms (AI)	Data structures & algorithms (AI)	Mathematical Economics 1
	Number Theory	Markov Chains	Markov Chains	Markov Chains	Markov Chains	Markov Chains	Markov Chains
	Measure Theory	Measure Theory	Measure Theory	Mech. And Term. of the cell	Equational Programming	Biomedical mathematics	Measure Theory
	Nonlinear Dynamical Systems	Nonlinear Dynamical	Nonlinear Dynamical Systems	Nonlinear Dynamical	Smart Systems Engineering	Information Retrieval	Nonlinear Dynamical Systems
	Project Computer Assisted Proof	Project Computer Assisted Proof	Project Computer Assisted Proof	Mathematics for Machine Learning	Project Computer Assisted Proof	Mathematics for Machine Learning	Project Computer Assisted Proof
Period 1-3, there may be clashes in the schedule		Biomedical mathematics		From protein to cell	Nonlinear Dynamical Systems	Data Analytics and Privacy	
				Metabolomics			
Periods 4-6, no clashes in the schedule	Differential Topology	Functional Analysis	Functional Analysis	Partial Differential	Databases	Mathematical Statistics 2	Mathematical Statistics 2
	Partial Differential Equations	Partial Differential Equations	Mathematical Statistics 2	Mathematical Statistics 2	Mathematical Statistics 2	Machine Learning	Applied Analysis: Financial Maths
	History of Science	History of Science	History of Science	History of Science	History of Science	History of Science	History of Science
	Philosophy and Ethics	Philosophy and Ethics	Philosophy and Ethics	Philosophy and Ethics	Philosophy and Ethics	Philosophy and Ethics	Philosophy and Ethics
	Bachelor project in Mathematics	Bachelor project in Mathematics	Bachelor project in Mathematics	Bachelor project in Mathematics	Bachelor project in Mathematics	Bachelor project in Mathematics	Bachelor project in Mathematics
Period 4-6, there may be clashes in the schedule	Functional Analysis	Applied Analysis: Financial Maths	Machine Learning	Machine Learning	Partial Differential Equations	Applied Analysis: Financial Maths	Mathematical Economics 2
			Applied Analysis: Financial Maths		Automata and Complexity	Complex Analysis	Operations Research 3
			Partial Differential Equations				

